

Glass Manufacturing Defects

Identification and Troubleshooting SOP

Glass Expert AI -- Sample SOP Document | Internal Reference

Section 1: Stones and Crystalline Inclusions

Stones are solid crystalline or amorphous inclusions in glass that did not dissolve during the melting process. They are one of the most serious defects in glass manufacturing because they can cause stress concentrations that lead to spontaneous breakage.

IDENTIFICATION: Stones appear as opaque or semi-opaque particles, typically white, grey, or colored, embedded within the glass. They can be identified by their sharp boundaries with the surrounding glass, their crystalline appearance under polarized light, and their chemical composition as determined by EDS or XRF analysis.

COMMON TYPES AND CAUSES:

1. Silica stones (cristobalite/quartz): Undissolved sand grains. Caused by insufficient melting temperature, too-coarse sand particle size, or insufficient melting time.
2. Refractory stones: Fragments from furnace refractory materials. Caused by refractory corrosion, thermal shock to refractory, or mechanical damage.
3. Devitrification stones: Crystalline phases that precipitate from the glass melt. Caused by contamination, incorrect composition, or slow cooling through the devitrification temperature range.
4. Batch stones: Incompletely reacted batch materials. Caused by poor batch mixing, incorrect batch composition, or insufficient melting.

CORRECTIVE ACTIONS:

- Increase melting temperature by 20-30 degrees C
- Extend melting time
- Check and correct batch composition and mixing
- Inspect and replace damaged refractory sections
- Review cullet quality and contamination sources

Section 2: Bubbles and Seeds

Bubbles (also called blisters or seeds when very small) are gas-filled voids in the glass. They range in size from less than 0.1 mm (seeds) to several millimeters (blisters). Bubbles are a common defect in glass manufacturing and can significantly impair optical and mechanical properties.

IDENTIFICATION: Bubbles appear as spherical or elongated voids. Seeds are very small bubbles, typically less than 0.5 mm in diameter. Blisters are larger bubbles, often found near the glass surface. The gas composition inside bubbles can be analyzed by mass spectrometry to determine their origin.

COMMON CAUSES AND GAS COMPOSITIONS:

1. SO₂/SO₃ bubbles: From decomposition of sulfate fining agents at high temperatures. Most common in soda-lime glass. Indicates insufficient fining.
2. CO₂ bubbles: From incomplete decomposition of carbonate batch materials (limestone, soda ash). Indicates insufficient melting temperature or time.
3. O₂ bubbles: From reboil of dissolved oxygen at forming temperatures. Common when glass is reheated after initial fining.
4. N₂ bubbles: From dissolved nitrogen in the melt, often from combustion atmosphere.
5. Water vapor bubbles: From moisture in batch materials or combustion gases.

FINING AGENTS AND THEIR MECHANISMS:

Fining agents are materials added to the batch to promote bubble removal. Common fining agents include:

- Sodium sulfate (Na₂SO₄): Releases SO₂/SO₃ at high temperature, creating large bubbles that rise quickly and carry smaller bubbles with them.
- Arsenic trioxide (As₂O₃): Releases O₂ during cooling, absorbing residual bubbles.
- Antimony trioxide (Sb₂O₃): Similar mechanism to arsenic, used as a safer alternative.
- Cerium oxide (CeO₂): Releases O₂ at fining temperatures.
- Tin oxide (SnO₂): Used in lead-free crystal glass.

2.1 Bubble Defect Troubleshooting Guide

Bubble Type	Gas Content	Location	Root Cause	Corrective Action
Seeds	SO ₂ /O ₂	Throughout	Insufficient fining	Increase fining agent, raise temp
Blisters	SO ₂	Surface	Reboil from sulfate	Reduce sulfate, check redox
CO ₂ seeds	CO ₂	Bottom	Incomplete melting	Increase melt temp/time
O ₂ bubbles	O ₂	Random	Reboil on reheating	Adjust forming temperature
Large bubbles	Mixed	Top	Refractory reaction	Inspect refractory

Section 3: Devitrification

Devitrification is the process by which a glass crystallizes, either during cooling from the melt or during reheating. It is a major quality problem in glass manufacturing because crystalline regions have different optical, mechanical, and thermal properties than the surrounding glass, leading to stress concentrations and potential breakage.

MECHANISM: Devitrification occurs in two stages: nucleation (formation of crystal nuclei) and crystal growth. The rate of both processes depends strongly on temperature. Nucleation is fastest at temperatures somewhat below T_g , while crystal growth is fastest at higher temperatures. The overall devitrification rate is the product of both processes and shows a maximum at an intermediate temperature.

For soda-lime-silica glass, the primary devitrification phase is devitrite ($\text{Na}_2\text{O} \cdot 3\text{CaO} \cdot 6\text{SiO}_2$), which forms as needle-like crystals typically at temperatures between 750 and 900 degrees Celsius. Cristobalite (SiO_2) can also form at higher temperatures in silica-rich compositions.

PREVENTION METHODS:

1. Composition control: Maintain composition within the stable glass-forming region. Avoid compositions near eutectic points in the phase diagram.
2. Nucleating agent control: Minimize the presence of nucleating agents such as TiO_2 , ZrO_2 , and P_2O_5 unless intentional (as in glass-ceramics).
3. Cooling rate control: Cool through the devitrification temperature range as quickly as possible to minimize residence time.
4. Surface cleanliness: Devitrification often initiates at the glass surface. Maintain clean forming tools and avoid contamination.
5. Redox control: Maintain appropriate redox conditions in the melt to prevent precipitation of crystalline phases.

Section 4: Cord, Striae, and Knots

Cord and striae are inhomogeneities in the glass that appear as thread-like or layered regions with a slightly different refractive index from the surrounding glass. They are caused by local variations in chemical composition within the glass melt.

IDENTIFICATION: Cord appears as visible threads or layers when the glass is viewed in transmitted light against a dark background. Striae are finer, more subtle variations that may require schlieren photography or interferometry to detect. Both defects cause optical distortion and are unacceptable in precision optical glass.

CAUSES:

1. Incomplete mixing of batch materials: Regions of different composition in the melt that have not been fully homogenized.
2. Refractory dissolution: Silica or alumina dissolved from furnace refractories creates high-viscosity regions that resist mixing.
3. Volatilization: Loss of volatile components (B_2O_3 , PbO , alkali oxides) from the melt surface creates a surface layer with different composition.
4. Cullet inhomogeneity: Use of cullet with different composition from the batch.

CORRECTIVE ACTIONS:

- Improve batch mixing and homogeneity
- Increase stirring in the refining zone
- Reduce refractory corrosion by adjusting melt chemistry
- Control volatilization by adjusting furnace atmosphere
- Ensure cullet composition matches batch composition

4.1 Summary of Glass Defect Types and Severity

Defect Type	Visual Appearance	Severity	Primary Cause
Stones	Opaque solid inclusions	Critical	Undissolved batch/refractory
Bubbles/Seeds	Spherical voids	High	Incomplete fining
Blisters	Large surface bubbles	High	Reboil/sulfate
Devitrification	Crystalline haze/crust	Critical	Composition/cooling
Cord	Visible threads/layers	Moderate-High	Compositional inhomogeneity
Striae	Fine optical distortion	Moderate	Minor comp. variation
Knots	Dense glassy inclusions	High	Refractory dissolution
Wires	Metallic inclusions	Critical	Metal contamination